

EE6320 RF Integrated Circuits
Project: PA Design

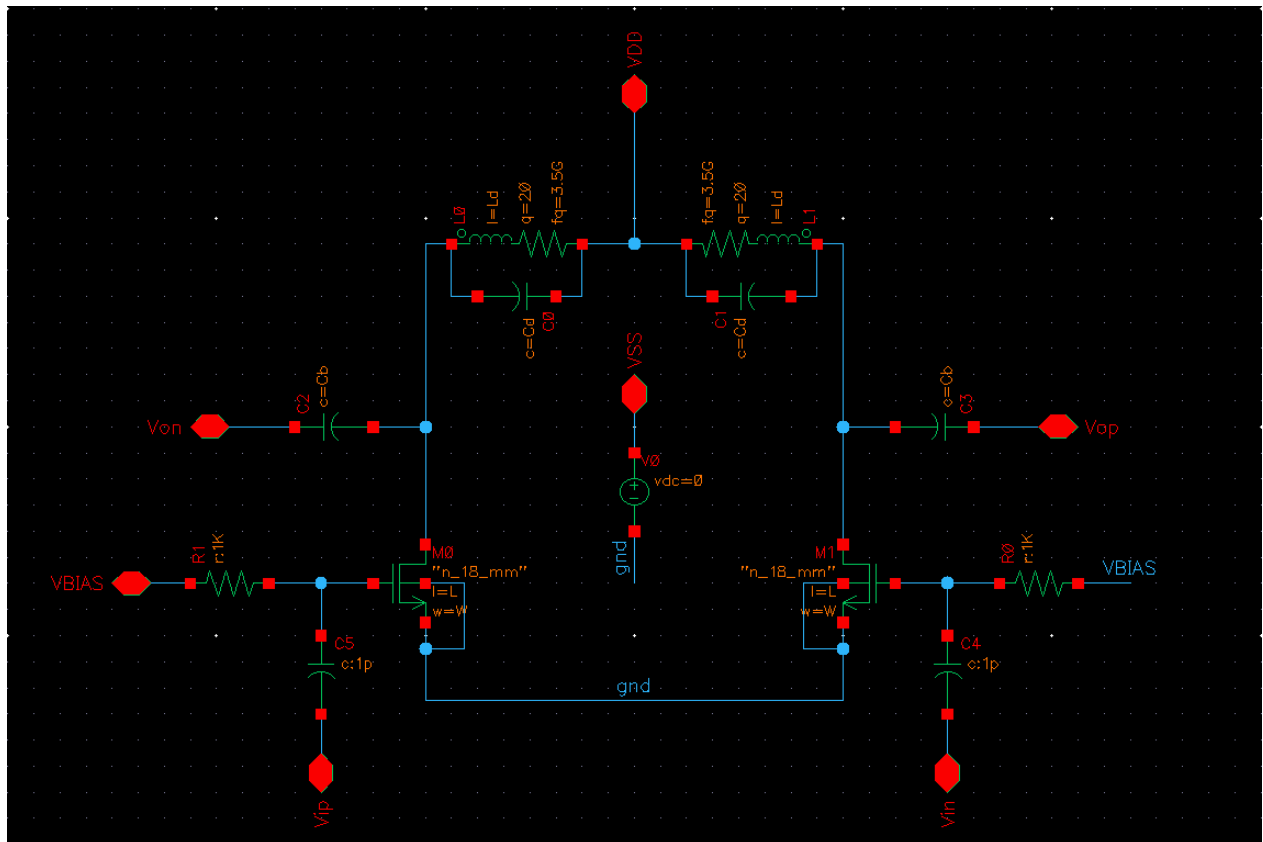
PA Performance Summary Table

Design Parameter	Design Metric	Performance	Specification
Output P1dB	$f_o = 3450$ MHz	13.4268dBm	$\geq +13$ dBm
	$f_o = 3500$ MHz	13.4156dBm	$\geq +13$ dBm
	$f_o = 3550$ MHz	13.4002dBm	$\geq +13$ dBm
AM-PM Deviation (at P1dB)	$f_o = 3450$ MHz	2.305 degrees	≤ 3 degrees
	$f_o = 3500$ MHz	2.00448 degrees	≤ 3 degrees
	$f_o = 3550$ MHz	1.70046 degrees	≤ 3 degrees
Voltage Gain (from Gate to Drain)	$f_o = 3450$ MHz	2.048	≥ 2
	$f_o = 3500$ MHz	2.055	≥ 2
	$f_o = 3550$ MHz	2.058	≥ 2
Power (at 3.5 GHz)	PA Average Consumption (excluding bias)	60.29244mW	Minimize
	Bias Circuit Consumption	3.6mW	Minimize
Other	Sum of all Capacitances (including ac coupling)	7.004nF	-
	Inductance Used	2nH	-

NAME : Venkata Subhash M
Roll No: NS25Z205

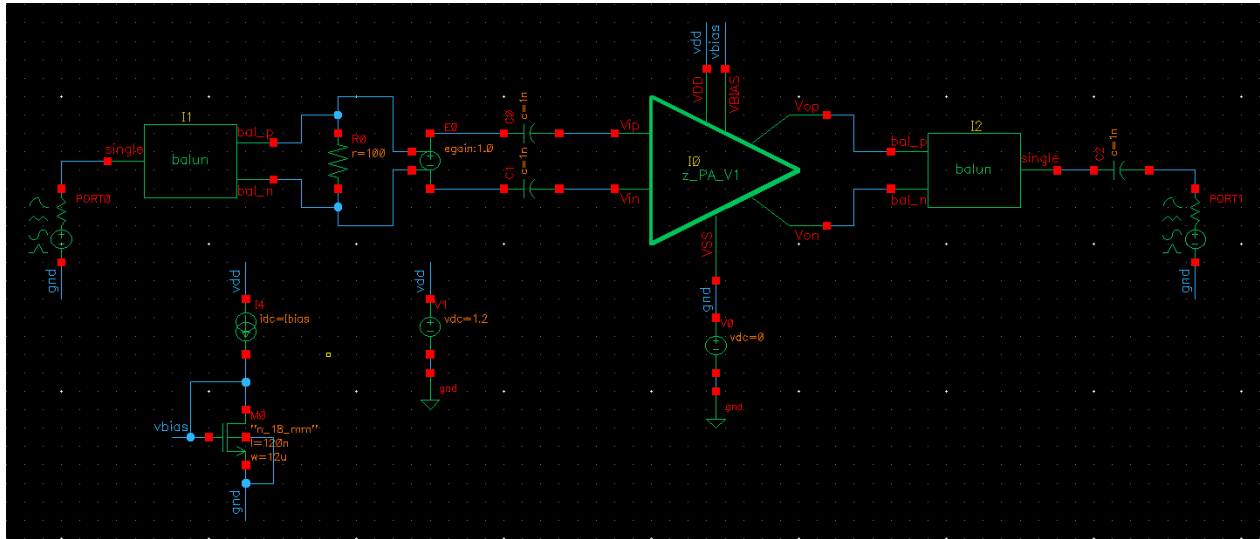
Device Width	120u	-
Simulator Used	virtuoso	-

PA Schematic



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PA Testbench

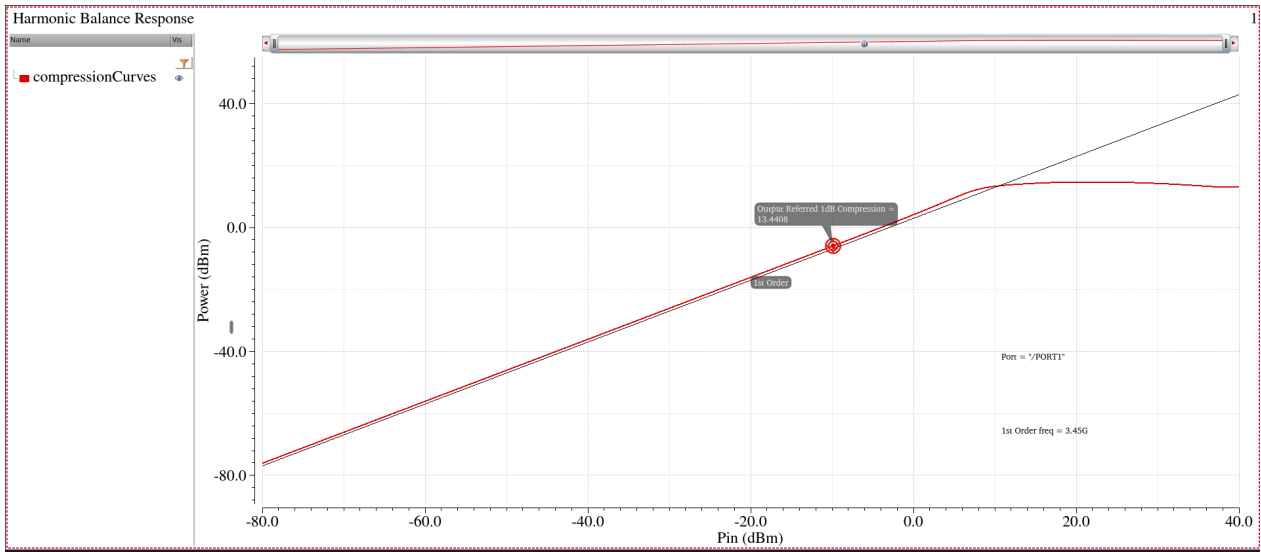


Component Values Table

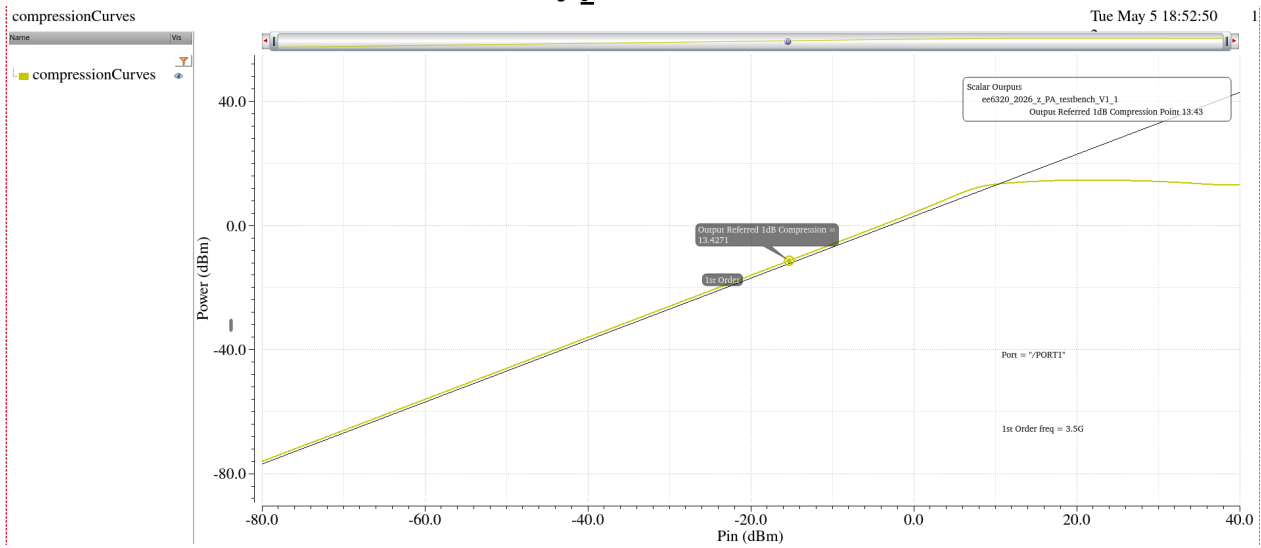
(W, L) MOS Amplifier = 120u,120n	(L, C) Drain Tank = 1nH,2pF
(W, L) Current Mirror MOS = 12u,120n	Bias Current = 3mA
$R_{bias} = 1\text{ k}\Omega$, $C_{coupling} = 1n$	$V_{DD} = 1.2$, $R_{load} = 50\text{ohm}$

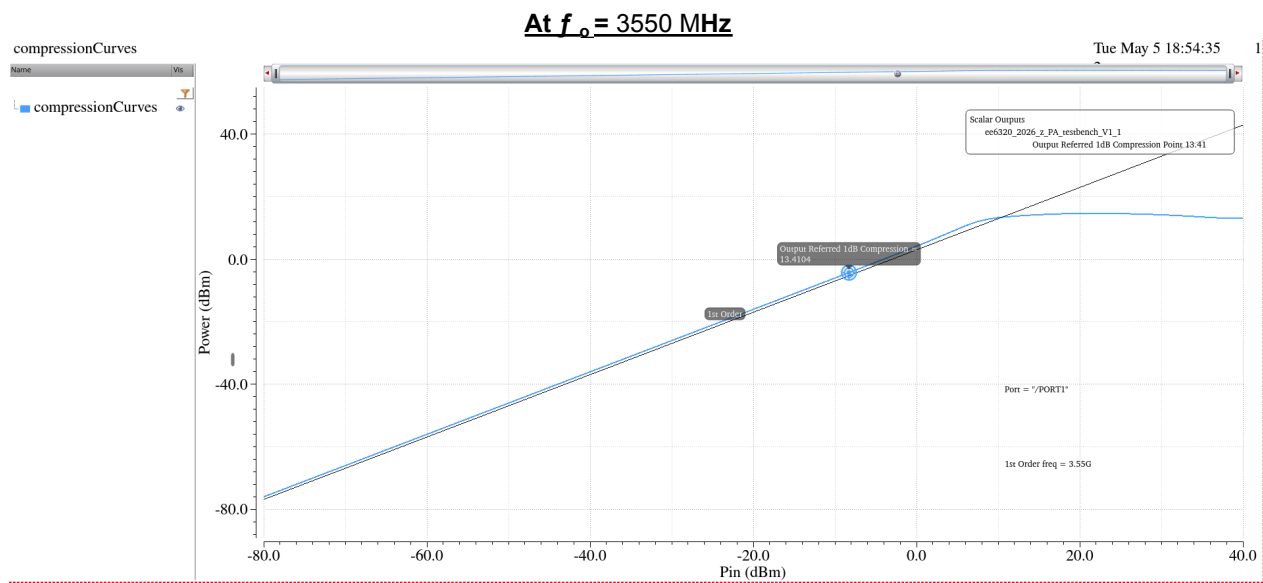
Output 1dB Compression Point Plots

At $f_o=3450$ MHz

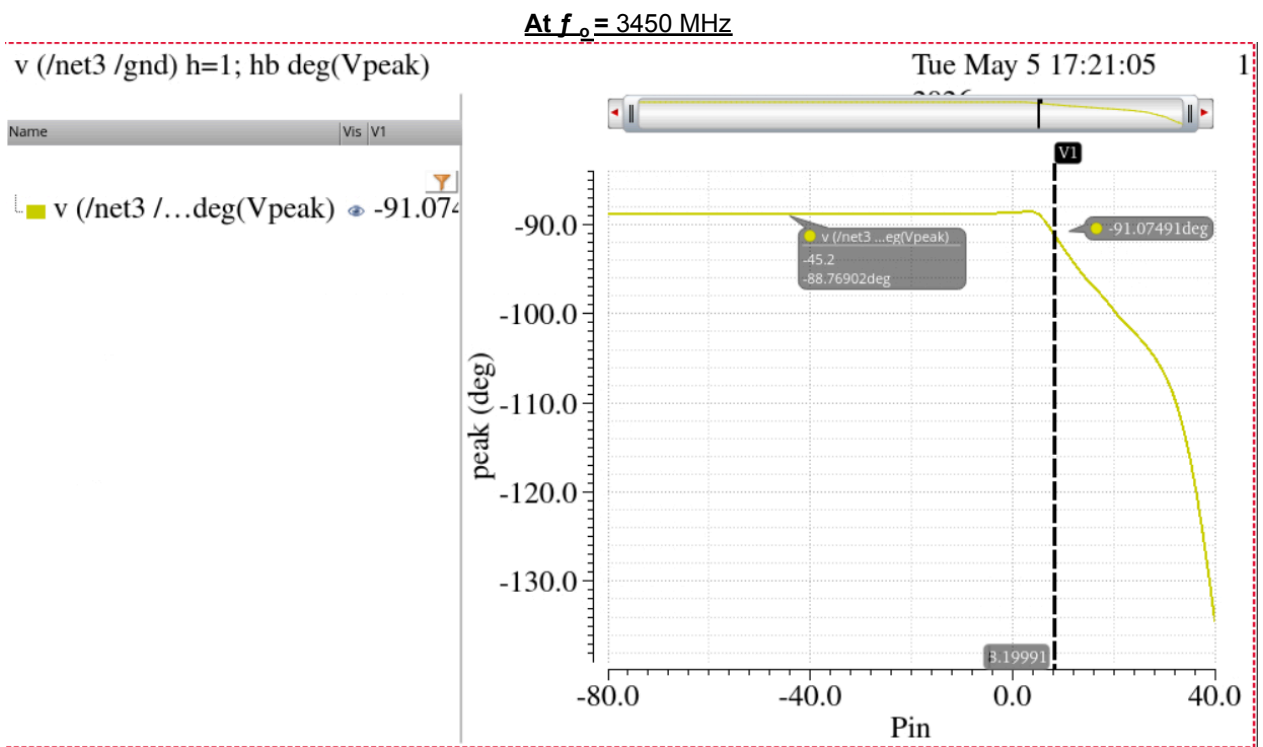


At $f_o=3500$ MHz

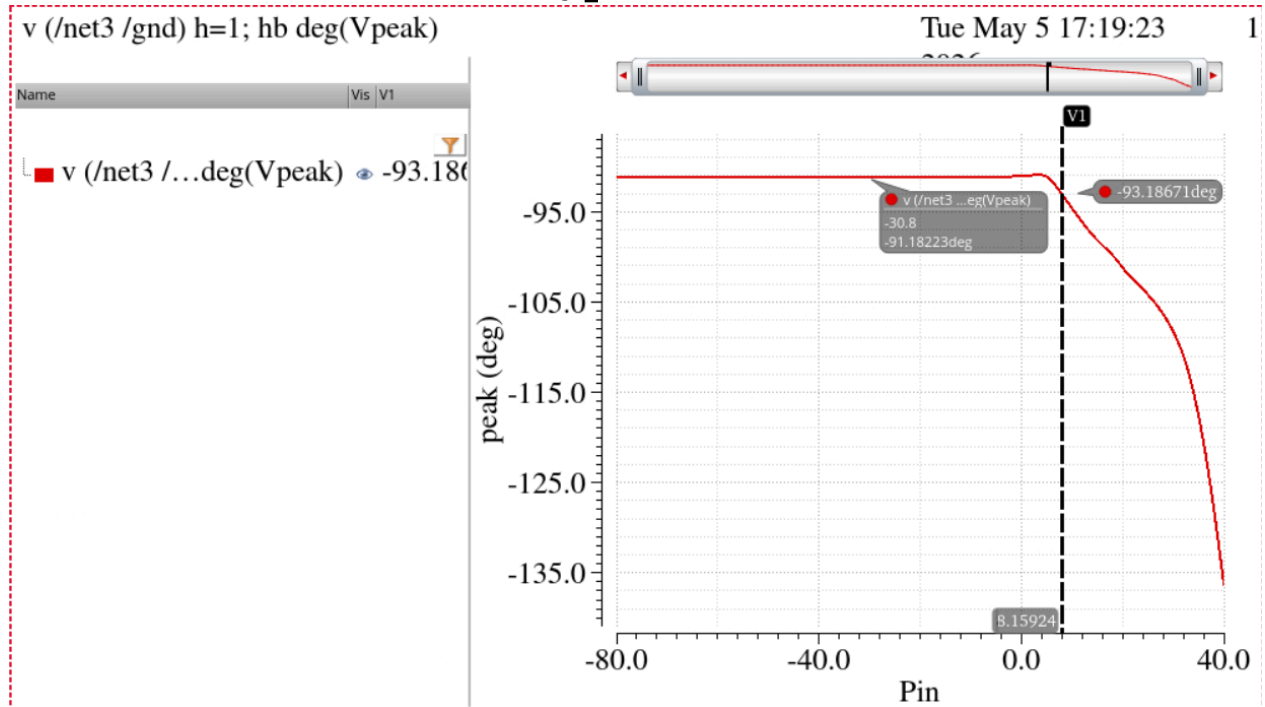




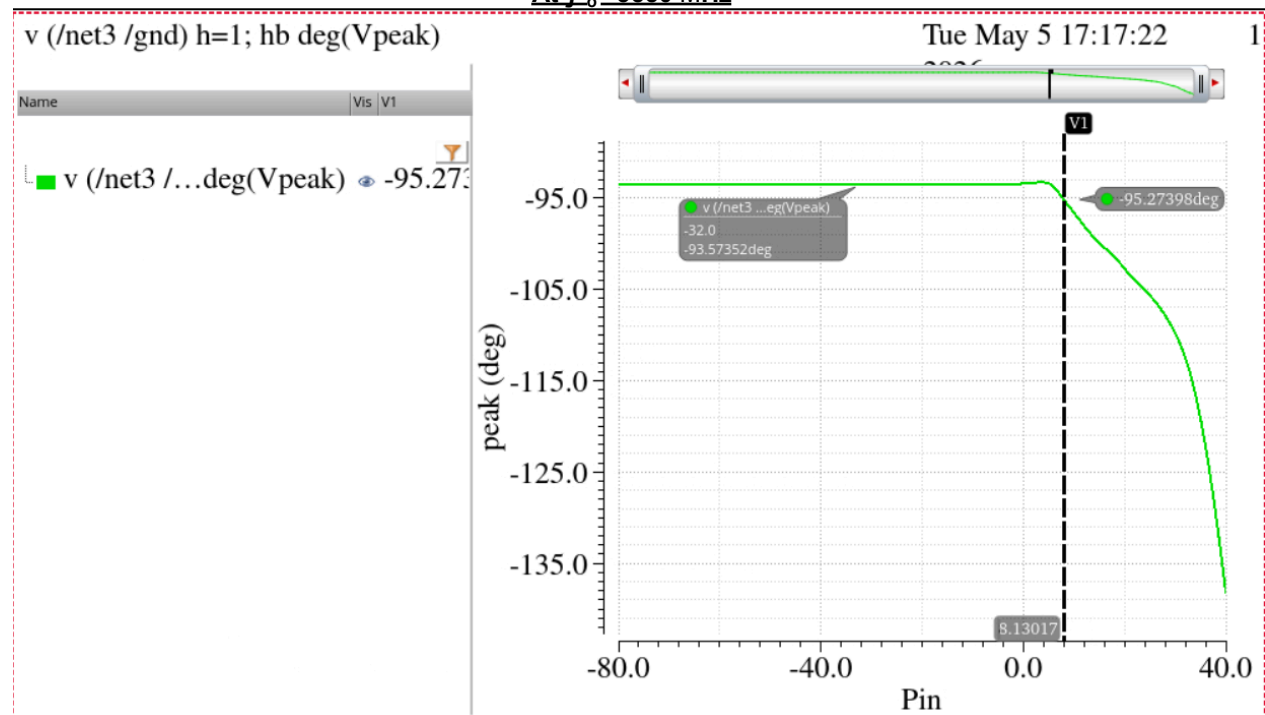
AM-PM Deviation Plots



At $f_o = 3500$ MHz

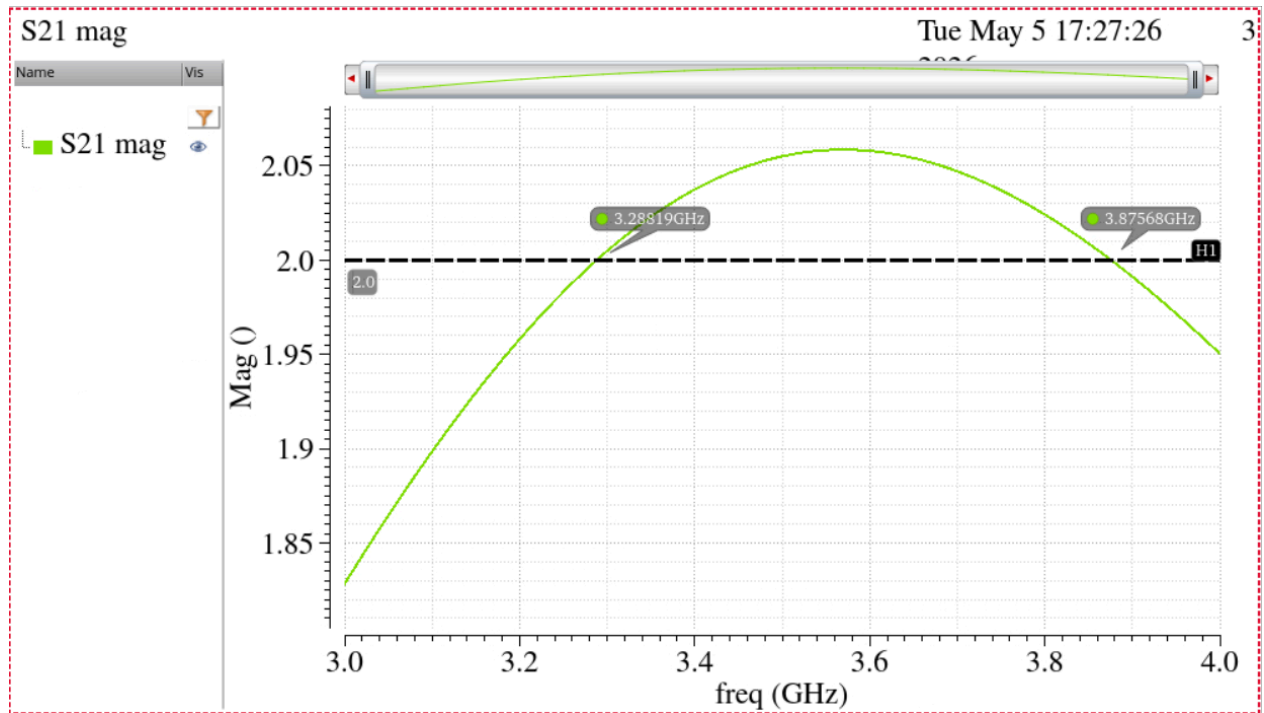


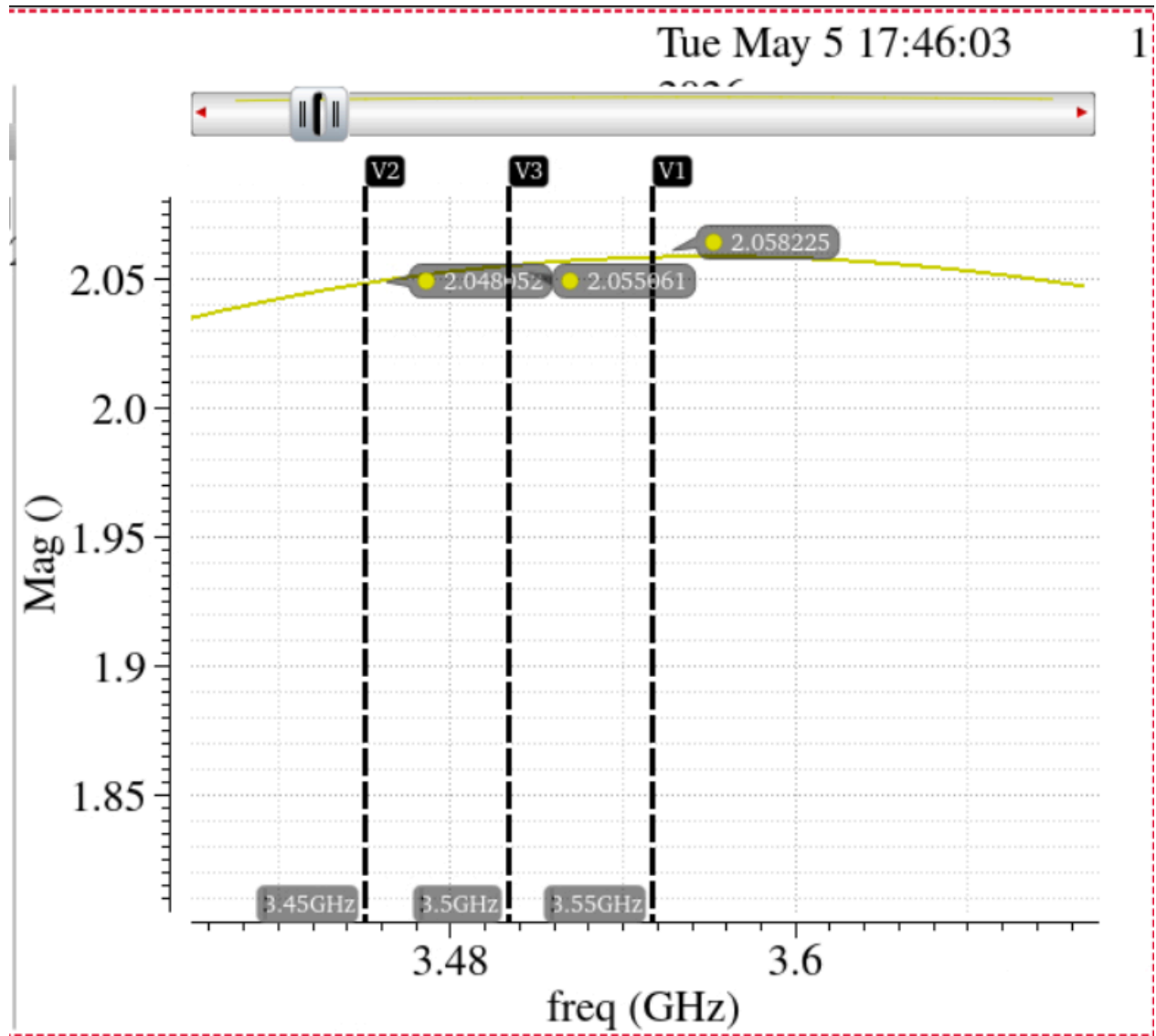
At $f_o = 3550$ MHz



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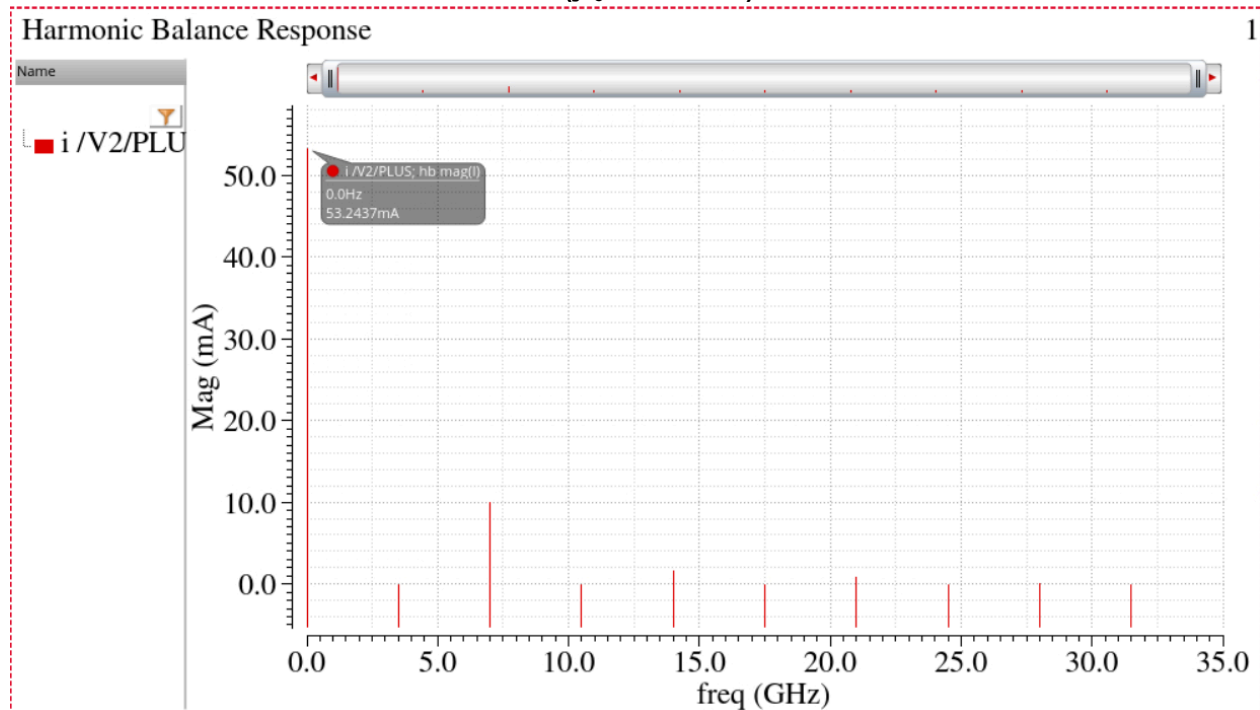
**PA Voltage Gain Plot
(from Gate to Drain)**





Current through VDD Plot

($f_o = 3500$ MHz)



Design Procedure

Fixed the inductor ,found a cap for 3.5G resonance and made the common source amplifier with nmos and sweep width and current until desired specs are achieved and changed the drain cap until I center the gain graph at 3.5G.

Path to Project Files

In demo library cell named PA have testbench

The symbol and schematic of it in there in RFIC 6320 library

Note: All the schematic/plots put up here are for reference only. Please replace them with the respective plots/schematic used or obtained by you.